Gurjot Singh Gaba, Ph.D.

Assistant Professor | Researcher | Young Professional Chair, IEEE Sweden

gurjot.singh@liu.se | gurjotsinghgaba | gurjotgaba.github.io

Gurjot Singh Gaba | 0000-0002-0732-1478 | +46-76-951-9925



Employment History

2026.02 – present

Researcher (Remote, Part-Time), Department of Computer Sciences, Kaunas University of Technology, Kaunas, Lithuania.

- *Conduct* research in cybersecurity focusing on post-quantum cryptography, quantum-ready infrastructures, and cross-border secure communication.
- *Develop* a quantum readiness framework for transitioning from classical to quantum-resilient network architectures.
- *Investigate* Cross-Border QKD, EuroQCI initiatives, and crypto-agility strategies for post-quantum adoption.
- *Design* and evaluate an EU PQC Readiness Framework, covering cryptographic inventory, risk assessment, migration planning, and governance.
- *Contribute* to an ontology-driven, context-aware scoring and reasoning framework for adaptive decision-making and evidence-based transition.

2025.07 – present

Assistant Professor of Cybersecurity, Department of Computer and Information Science (IDA), Linköping University, Linköping, Sweden.

- *Design* and deliver undergraduate and postgraduate modules covering foundational and advanced topics in cybersecurity.
- *Update* existing course curricula to align with the latest developments in cybersecurity research, industry standards, and regulatory frameworks.
- *Supervise* bachelor's, master's, and Ph.D. students on interdisciplinary research projects in cutting-edge areas of cybersecurity.
- *Coordinate* and contribute to EU Horizon and nationally funded research projects, fostering collaboration with academic and industrial partners.
- *Prepare* research grant proposals to secure external funding for expanding the AEGIS research group and advancing cybersecurity research.

2022.03 – 2025.06

Postdoctoral Researcher, Department of Computer and Information Science (IDA), Linköping University, Linköping, Sweden.

- *Lead* and manage research objectives and administrative tasks for ELLIIT (Sweden) and SESAR 3 JU (Horizon Europe) funded projects.
- *Conduct* vulnerability assessments and develop security solutions for cyber-physical systems in the industrial and aviation sectors.
- *Prepare* research grant proposals and manage intellectual property filings.
- *Supervise* PhD candidates and oversee their research progress.
- *Disseminate* research findings through conference papers, journal articles, blogs, seminars, and workshops.

Employment History (continued)

2021.09 – 2022.03 **Postdoctoral Researcher.** School of Computer and Communication Sciences, Mohammed VI Polytechnic University (UM6P), Benguerir, Morocco.

- *Conduct* cutting-edge research in cybersecurity, focusing on identifying emerging threats in the IoT and developing preventive strategies.
- *Publish* and present research findings through conference papers and peer-reviewed journal articles.
- *Write* and submit grant proposals to secure research funding.
- *Supervise* PhD and master's theses, offering mentorship and guidance.

2011.07 – 2021.08 **Assistant Professor.** School of Electronics and Electrical Engineering, Lovely Professional University (LPU), Punjab, India.

- *Serve* as faculty member, engaging in teaching and curriculum development.
- *Supervise* research projects and oversee the examination center as supt.
- *Coordinate* industrial training programs and manage guest lectures, seminars, and workshops.
- *Act* as alumni coordinator, maintaining strong relationships with graduates.
- *Manage* the admission process and coordinate accreditation efforts.

Education

2016 – 2021 **Ph.D. in Electronics and Electrical Engineering** with a *specialization in Cybersecurity* from Lovely Professional University, Punjab, India.
Thesis title: *Privacy-Preserving Authentication and Key Exchange Mechanisms in Internet of Things Applications.*

2013 – 2015 **Post Graduate Diploma in Business Administration** with a major in *Human Resource Management* from Symbiosis Centre for Distance Learning, Pune, India.

2009 – 2011 **M.Tech. in Electronics and Communication** *(Specialization in Wireless Communications)* from Guru Nanak Dev Engg. College, Punjab Technical University, Punjab, India.
Thesis title: *Network Layout Methodology for Wireless Sensor Network using Bacterial Foraging Optimization.*

2006 – 2009 **B.Tech. in Electronics and Communication** from Chandigarh Engineering College, Punjab Technical University, Punjab, India.

Funded Research Projects & Grants

2026–2027

Visual Sensing and Decision Support for Sustainable Food Redistribution: A Swedish Pilot within the FEAST Framework | *Co-Principal Investigator (Co-PI)*
Funded by Visual Digital Futures, Linköping University | *Total Allocation: SEK 0.388 M*
Contribution: Leading the cybersecurity research within an interdisciplinary project exploring trustworthy digital technologies for sustainable food redistribution. My work focuses on:

- Analysing the digital food redistribution ecosystem, including stakeholder interactions, information flows, critical assets, and trust boundaries, to identify cybersecurity threats and vulnerabilities.
- Conducting systematic threat modelling and risk assessment to identify and prioritise security, privacy, and trust challenges in multi-stakeholder food redistribution systems.
- Deriving security, privacy, and trust requirements and developing a secure reference architecture for trustworthy information sharing across visual sensing, IoT, and decision-support systems.
- Evaluating the proposed framework through threat–requirement–architecture traceability analysis and contributing cybersecurity expertise to the Swedish pilot and future international research collaborations.

Key Skills: Cybersecurity Threat Modelling, Risk Assessment, Security and Privacy Requirements, Trustworthy Digital Infrastructures, and Interdisciplinary Research.

2022–2027

Digital4Business – EU-Funded Research Project
Funded by the EC under the Digital Europe Programme | *Total Funding: EUR 19.92 M*
Contribution: Involved in the design, academic delivery, and continuous development of the *Cybersecurity curriculum* for Digital4Business, a European master's program aimed at enhancing digital skills for business professionals. My role involves:

- Designing, delivering, and continuously updating course content, including lectures, presentations, case studies, and educational videos.
- Identifying key cybersecurity topics relevant to business professionals.
- Ensuring curriculum alignment with best practices from leading universities.
- Overseeing the progress of the module to ensure its alignment with project goals.
- Participating in Digital4Business consortium meetings (15 partners, 7 countries) to present updates and contribute to discussions on digital education in Europe.

Key Skills: Teaching, Course Development, Project Management, Cybersecurity.

Funded Research Projects & Grants (continued)

2023–2026 **SEC-AIRSPACE - Cyber Security Risk Assessment in virtualized AIRSPACE scenarios and stakeholders' awareness of building resilient ATM** | *Co-PI*
Funded by the EU SESAR Joint Undertaking | *Total Funding: EUR 1 M*
Contribution: Leading cybersecurity research within SEC-AIRSPACE, a project addressing cyber threats in both legacy and future communication infrastructures (FCI) of aviation. My role focused on:

- Identified potential cyber threats and vulnerabilities in aviation's FCI.
- Evaluated the SESAR-introduced cyber risk assessment methodology (SecRAM) for effectiveness, scope, and coverage.
- Conducted validation exercises by exposing Air Traffic Controllers to simulated cyber-attacks and proposed measures to enhance their cyber awareness.
- Managed project deliverables and presented progress to consortium members (9 partners from 5 countries).
- Contributed to proposal writing and led validation exercises as Task Leader.

Key Skills: Cyber Threat & Vulnerability Discovery, Cyber Risk Assessment, Validation Activities, Proposal Development.

2022–2023 **Secure Transparent Communications in the Industrial Internet** | *Researcher*
Funded by ELLIIT Excellence Center at Linköping–Lund in Information Technology
Contribution: Conducted research on securing Industrial Internet communication infrastructures, focusing on cybersecurity, privacy, and resilient industrial networking. My role focused on:

- Investigated cybersecurity and privacy challenges across the Industrial Internet, Industry 4.0, and emerging Industry 5.0 ecosystems, with emphasis on cyber-physical systems and IT/OT convergence.
- Developed threat models and analysed cyber risks, attack vectors, vulnerabilities, and security architectures to improve the resilience of industrial communication systems.
- Explored AI-driven, blockchain-enabled, edge-computing, and privacy-preserving security mechanisms to strengthen secure and transparent industrial communications.

Key Skills: Cyber-Physical Systems, Threat Modelling, Privacy.

Additional Grants

2026–2028 **CyberResilient-Nordic: A Cross-Sector Nordic Research Network on Cyber Resilience and Digital Sovereignty in Critical Infrastructures** | *National Coordinator & WP Leader*
Funded by NordForsk | *Total Allocation: NOK 2.0 M*
Contribution: Leading Sweden's participation in CyberResilient-Nordic, a Nordic research network advancing cyber resilience and digital sovereignty across aviation, energy, and maritime critical infrastructures. My role focuses on:

- Coordinating Sweden's participation as National Coordinator while leading WP3 (Summer Schools and Capture-the-Flag Competitions) to strengthen cybersecurity education, interdisciplinary training, and capacity building for early-career researchers and PhD students.
- Establishing cross-sector collaboration among academia, industry, and critical infrastructure stakeholders through researcher exchanges, industrial workshops, and Nordic research networking to advance cyber resilience and digital sovereignty.
- Contributing to the development of shared training resources, strategic research roadmaps, and competitive Nordic and European research proposals in cybersecurity for critical infrastructures.

Key Skills: Cyber Resilience, Digital Sovereignty, International Research Leadership.

2026–2027 **Networked Hubs of Competencies for Emerging Technologies-Driven Cyber-Sustainable Engineering** | *National Lead*
Funded by Nordplus Higher Education (NPHE-2026/10449) | *Total Allocation: EUR 0.061 M*
Contribution: Leading Linköping University's participation within the ENERGYCOM network to strengthen Nordic–Baltic engineering education through emerging technologies, cyber-sustainability, and international collaboration. My role focuses on:

- Leading Linköping University's activities within the Sustainable Communications Hub, contributing expertise in wireless networking, network security, data privacy, and secure communication infrastructures.
- Coordinating national project implementation, supporting international mobility, joint workshops, shared laboratories, and cross-institutional educational activities across Nordic–Baltic partner universities.
- Contributing to project governance, dissemination, reporting, and the integration of AI, cyber-physical systems, digital twins, and other emerging technologies into cyber-sustainable engineering education.

Key Skills: Project Leadership, Cybersecurity Education, Academic Collaboration.

Additional Grants (continued)

2025–2026

Capture the Flag (CTF) Cybersecurity Challenge 2025 | Lead Organiser
Funded by the ELLIIT | *Event Budget*: SEK 90 000
Contribution: Led the design, organisation, and execution of the 2025 CTF event, an initiative aimed at addressing the national shortage of cybersecurity professionals and encouraging greater participation of women in the field. The event brought together over one hundred participants across educational and professional levels and provided hands-on exposure to real-world cybersecurity challenges. The scope of responsibilities included:

- Directed full event planning and execution, including challenge design coordination, logistics, budget allocation, and stakeholder engagement.
- Strengthened diversity and inclusion in cybersecurity by integrating female keynote speakers, women-designed challenges, and targeted outreach activities.
- Managed collaboration with academic, industry, and student organisations, including technical support partners and advisory cybersecurity experts.
- Oversaw reporting, impact assessment, and dissemination of outcomes to ensure continuity and improvement for future editions.

Key Skills: Cybersecurity Event Leadership, Stakeholder and Community Engagement, Strategic Planning and Execution.

Advancing the Education of Engineers Through AI for Cyber-Sustainable and Energy-Efficient Union | *National Lead*
Funded by Nordplus Higher Education (NPHE-2025/10355) | *Total Allocation: EUR 0.128 M*
Contribution: Led national responsibilities within the ENERGYCOM network to modernise engineering education through AI-enhanced, cyber-sustainable teaching practices. My role included:

- Oversaw national project implementation, ensuring financial accountability, compliance, and timely delivery of project activities.
- Coordinated international collaboration to support mobility programmes, workshops, and capacity-building activities among Nordic–Baltic partner institutions.
- Integrated AI-powered educational tools and cybersecurity-oriented pedagogical approaches to strengthen digital competence, student engagement, and engineering education.
- Led dissemination activities and contributed to project reporting, impact assessment, and the preparation of technical and administrative deliverables.

Key Skills: AI-Enhanced Learning, Cybersecurity Education, Academic Project Management, International Collaboration.

Research Publications

Journal Articles

1. M. Borhani, **G. S. Gaba**, D. Kuptsov, and A. Gurtov, "Towards secure and scalable provider provisioned layer 2 VPNs: Embracing zero trust approach," *IEEE Transactions on Consumer Electronics*, pp. 1–12, 2026.

2 **G. S. Gaba**, A. Sari, I. Butun, P. Singh, A. Gurtov, and M. Liyanage, "A survey on security and privacy of industry 4.0 and beyond: Technical aspects, use cases, challenges, and research directions," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 8865–8929, 2025.

3 M. Borhani, **G. S. Gaba**, J. Basaez, I. Avgouleas, and A. Gurtov, "A critical analysis of the industrial device scanners' potentials, risks, and preventives," *Journal of Industrial Information Integration*, vol. 41, p. 100 623, 2024.

4 S. Khan, **G. S. Gaba**, A. Gurtov, L. J. Jansen, N. Mäurer, and C. Schmitt, "Post quantum secure handover mechanism for next generation aviation communication networks," *IEEE Transactions on Green Communications and Networking*, vol. 8, no. 3, pp. 939–955, 2024.

5 A. Rehaimi, Y. Sadqi, Y. Maleh, **G. S. Gaba**, and A. Gurtov, "Towards a federated and hybrid cloud computing environment for sustainable and effective provisioning of cyber security virtual laboratories," *Expert Systems With Applications*, vol. 252, no. Part B, p. 124 267, 2024.

6 K. Zerhouni, **G. S. Gaba**, M. Hedabou, T. Maksymyuk, A. Gurtov, and E. M. Amhoud, "GAN-based evasion attack in filtered multicarrier waveforms systems," *IEEE Transactions on Machine Learning in Communications and Networking*, vol. 2, pp. 210–220, 2024.

7 A. Zineddine, O. Chakir, **G. S. Gaba**, *et al.*, "A systematic review of cybersecurity assessment methods for HTTPS," *Computers and Electrical Engineering*, vol. 115, no. April 2024, p. 109 137, 2024.

8 O. Chakir, A. Rehaimi, Y. Sadqi, M. Krichen, **G. S. Gaba**, A. Gurtov, *et al.*, "An empirical assessment of ensemble methods and traditional machine learning techniques for web-based attack detection in industry 5.0," *Journal of King Saud University-Computer and Information Sciences*, vol. 35, no. 3, pp. 103–119, 2023.

9 S. Khan, **G. S. Gaba**, A. Braeken, P. Kumar, and A. Gurtov, "AKAASH: A realizable authentication, key agreement, and secure handover approach for controller-pilot data link communications," *International Journal of Critical Infrastructure Protection*, vol. 42, p. 100 619, 2023.

10 **G. S. Gaba**, M. Hedabou, P. Kumar, A. Braeken, M. Liyanage, and M. Alazab, "Zero knowledge proofs based authenticated key agreement protocol for sustainable healthcare," *Sustainable Cities and Society*, p. 103 766, 2022.

11 **G. S. Gaba**, M. Masud, P. Kumar, and A. Gurtov, "A user-centric privacy-preserving authentication protocol for IoT-AmI environments," *Computer Communications*, vol. 196, pp. 45–54, 2022.

12 M. Masud, **G. S. Gaba**, K. Choudhary, M. S. Hossain, M. F. Alhamid, and G. Muhammad, "Lightweight and anonymity-preserving user authentication scheme for IoT-based healthcare," *IEEE Internet of Things Journal*, vol. 9, no. 4, pp. 2649–2656, 2022.

13 P. Singh, **G. S. Gaba**, A. Kaur, M. Hedabou, and A. Gurtov, "Dew-cloud-based hierarchical federated learning for intrusion detection in IoMT," *IEEE journal of biomedical and health informatics*, vol. 27, no. 2, pp. 722–731, 2022.

14 C. Boguszewicz, M. Boguszewicz, **G. S. Gaba**, *et al.*, "The fourth industrial revolution – cyberspace mental wellbeing: Harnessing science & technology for humanity (white paper)," *Global foundation for cyber studies and research*, 2021.

15 **G. S. Gaba**, G. Kumar, T.-H. Kim, H. Monga, and P. Kumar, "Secure device-to-device communications for 5G enabled internet of things applications," *Computer Communications*, vol. 169, pp. 114–128, 2021.

16 H. S. Gill, T. Singh, B. Kaur, **G. S. Gaba**, M. Masud, and M. Baz, "A metaheuristic approach to secure multimedia big data for IoT-based smart city applications," *Wireless Communications and Mobile Computing*, vol. 2021, no. 1, p. 7 147 940, 2021.

17 M. Masud, M. Alazab, K. Choudhary, and **G. S. Gaba**, "3P-sake: Privacy-preserving and physically secured authenticated key establishment protocol for wireless industrial networks," *Computer Communications*, vol. 175, pp. 82–90, 2021.

18 M. Masud, **G. S. Gaba**, K. Choudhary, R. Alroobaea, and M. S. Hossain, "A robust and lightweight secure access scheme for cloud based e-healthcare services," *Peer-to-peer Networking and Applications*, vol. 14, no. 5, pp. 3043–3057, 2021.

19 K. Choudhary, **G. S. Gaba**, I. Butun, and P. Kumar, "MAKE-IT—a lightweight mutual authentication and key exchange protocol for industrial internet of things," *Sensors*, vol. 20, no. 18, p. 5166, 2020.

20 **G. S. Gaba**, G. Kumar, H. Monga, T.-H. Kim, and P. Kumar, "Robust and lightweight mutual authentication scheme in distributed smart environments," *IEEE Access*, vol. 8, pp. 69 722–69 733, 2020.

21 **G. S. Gaba**, G. Kumar, H. Monga, T.-H. Kim, M. Liyanage, and P. Kumar, "Robust and lightweight key exchange (LKE) protocol for industry 4.0," *IEEE Access*, vol. 8, pp. 132 808–132 824, 2020.

22 M. Masud, **G. S. Gaba**, S. Alqahtani, *et al.*, "A lightweight and robust secure key establishment protocol for internet of medical things in COVID-19 patients care," *IEEE Internet of Things Journal*, vol. 8, no. 21, pp. 15 694–15 703, 2020.

23 V. Siddhartha, **G. S. Gaba**, and L. Kansal, "A lightweight authentication protocol using implicit certificates for securing IoT systems," *Procedia Computer Science*, vol. 167, pp. 85–96, 2020.

Conference Proceedings

1 M. Hagl, S. Boonsong, T. H. Stelkens-Kobsch, T. Ruediger, A. Gurtov, and **G. S. Gaba**, "Fake aircraft, real threats: Training air traffic controllers for cyberattacks," in *Proceedings of the 9th International Conference on Intelligent Human Systems Integration (IHSI 2026): Disruptive and Innovative Technologies*, vol. 200, Florence, Italy: AHFE International, 2026.

2 S. Boonsong, M. Hagl, T.-H. Stelkens-Kobsch, T. Ruediger, A. Gurtov, and **G. S. Gaba**, "Validation exercise: Strengthening cybersecurity awareness in air traffic control through simulation-based training," Poster presented at SESAR Innovation Days (SIDs) 2025, Lake Bled, Slovenia, Dec. 2025.

3 D. Martintoni, V. Senni, **G. S. Gaba**, and A. Gurtov, "On the role of SecRAM catalogues in ATM cyber risk assessment and improvement opportunities," in *Integrated CNS: Towards Innovative and Efficient CNS Service Provision*, IEEE, EUROCONTROL Brussels Headquarters, Belgium, 2025, pp. 1–8.

4 A. Rosén, **G. S. Gaba**, and A. Gurtov, "A strategic roadmap for phased zero trust architecture implementation in organizations," in *IEEE Conference on Communications and Network Security (CNS)*, Avignon, France, 2025, pp. 1–6.

5 T. H. Stelkens-Kobsch, M. Hagl, **G. S. Gaba**, *et al.*, "Air traffic management security in the digital age: A proactive risk management strategy," in *2025 AIAA DATC/IEEE 44th Digital Avionics Systems Conference (DASC)*, Montreal, QC, Canada, 2025, pp. 1–10.

6 K. Bernsmed, P. H. Meland, **G. S. Gaba**, *et al.*, "SEC-AIRSPACE: Addressing cyber security challenges in future air traffic management," in *SECURWARE 2024: The Eighteenth International Conference on Emerging Security Information, Systems and Technologies*, Nice, France, 2024, pp. 165–171.

7 C. Jensen, M. Karresand, **G. S. Gaba**, A. Gurtov, and E. Öhrn, "Are swedish passwords tougher than the rest?" In *Nordic Conference on Secure IT Systems*, Springer, Karlstad, Sweden, 2024, pp. 3–21.

8 S. Khan, **G. S. Gaba**, F. Boeira, and A. Gurtov, "Formal verification and security assessment of the drone remote identification protocol," in *2nd International Conference on Unmanned Vehicle Systems-Oman (UVS)*, IEEE, Muscat, Oman, 2024, pp. 1–8.

9 S. Khan, Y. Wang, **G. S. Gaba**, A. Gurtov, and P. Kumar, "A secure framework for controller pilot data link communications in aviation network," in *AIAA DATC/IEEE 43rd Digital Avionics Systems Conference (DASC)*, IEEE, San Diego, USA, 2024, pp. 1–11.

10. S. Khan, **G. S. Gaba**, A. Gurtov, N. Mäurer, T. Gräupl, and C. Schmitt, "Enhancing cybersecurity for LDACS: A secure and lightweight mutual authentication and key agreement protocol," in *2023 IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC)*, IEEE, Barcelona, Spain, 2023, pp. 1–10.

11. L. Nee, **G. S. Gaba**, and A. Gurtov, "Peer-to-peer misbehavior reporting using non-interactive zero-knowledge proofs for intelligent transport systems," in *IEEE CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON)*, IEEE, 2023, pp. 1–6.

12. S. Khan, **G. S. Gaba**, and A. Gurtov, "A federated learning based security for controller pilot data link communication," in *33rd Congress of the International Council of the Aeronautical Sciences, Stockholm, Sweden*, 2022, pp. 1–13.

Books and Chapters

1. E. Konstantopoulou, **G. S. Gaba**, and N. Sklavos, "AI and Blockchain for secure 6G wireless communications: Challenges, opportunities, and research directions," in *AI and Blockchain for Wireless Communications: Creating Resilient and Secure Systems*, F. Ayaz and Z. Sheng, Eds., Elsevier, 2026, pp. 237–254.

2. K. Choudhary, **G. S. Gaba**, R. Miglani, L. Kansal, and P. Kumar, "Artificial intelligence and machine learning aided blockchain systems to address security vulnerabilities and threats in the industrial internet of things," in *Intelligent Wireless Communications*, IET, 2021, pp. 329–361.

3. P. Kumar and **G. S. Gaba**, "Biometric-based robust access control model for industrial internet of things applications," in *IoT Security: Advances in Authentication*, John Wiley & Sons, 2019, pp. 133–142.

Granted Patents

1. **G. S. Gaba**, P. K. Raghav, V. Singh, and D. S. Bisht, *Method for preventing baggage theft and facilitating hassle free collection of baggage at airports*, IN, Patent No. 539946, 2024.

2. **G. S. Gaba**, S. G. Padder, M. S. Singh, and N. Rai, *An automatic shielding apparatus for protecting a vehicle*, IN, Patent No. 370963, 2021.

3. **G. S. Gaba**, S. A. Verma, K. K. Jain, P. Badala, and S. Krishnaa, *Novel seat locator system for theaters and auditoriums*, IN, Patent No. 377587, 2021.

Honors and Awards

2025 — **Distinguished Teacher Award** in the category of *Student's Hero*, presented by the Department of Computer and Information Science (IDA), Linköping University, Sweden.

Best Paper Award, The 25th Integrated Communications, Navigation, and Surveillance Conference (ICNS), Eurocontrol Headquarters, Brussels, Belgium, April 8–10.

2019 — **Research Appreciation Award**, presented by Lovely Professional University, India.

2016 — **Distinguished Teacher Award**, presented by Lovely Professional University, India, in a ceremony officiated by Smriti Zubin Irani, then Union Minister of Human Resource Development, Government of India.

Certifications

2026-07 — **International English Language Testing System (IELTS) – General Training** (Overall Band 8.0, CEFR C1), jointly issued by the British Council, IDP, and Cambridge English; demonstrating advanced English proficiency across listening (6.5), reading (9.0), writing (7.0), and speaking (8.5).

Certifications (continued)

2026-04 **Certified in Cybersecurity (CC)**, issued by ISC2 (International Information System Security Certification Consortium); ANAB-accredited (ISO/IEC 17024) credential demonstrating foundational competencies in security principles, network security, access control, security operations, and incident response.

2026-01 **Higher Education Pedagogy for Teachers Responsible for Courses** (120 hours / 4.5 ECTS), awarded by Didacticum, Linköping University, Sweden.

2025-10 **Examiner in Higher Education - Role and Responsibilities** (20 hours / 0.75 ECTS), awarded by Didacticum, Linköping University, Sweden.

2025-07 **Higher Education Pedagogy for New Teachers** (120 hours / 4.5 ECTS), awarded by Didacticum, Linköping University, Sweden.

Supervision of Independent (Degree) Projects (80 hours / 3 ECTS), awarded by Didacticum, Linköping University, Sweden.

Teaching Experience

Lecture Courses

2026–present **Information Security: Privacy, System and Network Security (TDDE62)**, Linköping University, Sweden.

Databases and Information Security for Bioinformatics (TDDE49), Linköping University, Sweden.

2018-2021 **Cryptography & Network Security (CSE 403)**, Lovely Professional University, India.

2016-2021 **Computer Networks (CSE 306)**, Lovely Professional University, India.

Laboratory Instruction

2023-2026 **Computer Networks and Internet Protocols (TDTS11)**, Linköping University.

2022-2025 **Computer Networks (TDTS06)**, Linköping University, Sweden.

Research and Professional Engagements

2026.09 – present **Management Committee Member,** Secure and Adaptive Frameworks for Environmentally sustainable Offshore WIND expansion (SAFEWIND) (CA25138), COST Association - European Cooperation in Science and Technology, Brussels, Belgium.

- *Contribute* cybersecurity expertise to the identification and assessment of threats affecting AI-enabled offshore wind infrastructure.
- *Support* research activities addressing the security, resilience, and trustworthiness of data and AI integration in offshore wind energy systems.

2026.10 – present **Management Committee Member,** Moving Data for Europe (MoveDataMove) (CA25140), COST Association - European Cooperation in Science and Technology, Brussels, Belgium.

- *Bring* cybersecurity and data-protection perspectives to research on user-driven data access and portability across European data ecosystems.
- *Examine* security and trust considerations associated with mechanisms for data access, sharing, and reuse.

Research and Professional Engagements (continued)

2025.03 – 2026.12 **Young Professional Chair,** IEEE Sweden Section.

- *Organize* professional development programs, networking events, and mentorship initiatives for Young Professionals in Sweden.
- *Represent* the IEEE Sweden Section in regional and international meetings, fostering collaborations and engagement within the global IEEE community.

2024.08 – 2025.07 **Visiting Researcher,** Department of Computer Science, Warwick Manufacturing Group (WMG), Warwick University, Warwick, United Kingdom.

- *Evaluate* vulnerabilities in the Host Identity Protocol (HIP) by conducting in-depth security assessments.
- *Perform* comprehensive vulnerability analyses to identify security weaknesses and assess their potential impact.
- *Develop* and implement countermeasures to enhance the security and resilience of next-generation networks.
- *Present* findings and recommendations to key stakeholders, including academic and industry partners.

2023.09 – present **Management Committee Member,** Behavioral Next Generation in Wireless Networks for Cyber Security (CA22104), COST Association - European Cooperation in Science and Technology, Brussels, Belgium.

- *Contribute* to strategic decision-making and policy development within the COST Action to support international research collaboration.
- *Participate* in meetings, working groups, and discussions to shape research directions, funding allocations, and networking activities.
- *Facilitate* knowledge exchange and interdisciplinary cooperation by connecting researchers, institutions, and industry stakeholders across Europe.

2023.03 – 2025.12 **Secretary,** IEEE Sweden Section.

- *Coordinate* administrative activities, maintain records, and ensure effective communication within the IEEE Sweden Section.
- *Represent* the IEEE Sweden Section in regional and international meetings, fostering collaborations and engagement within the global IEEE community.

2021.06 – 2022.05 **Researcher (Honorary Position),** Global Foundation for Cyber Studies and Research, Washington, United States.

- *Conduct* research, analyze cybersecurity trends, and publish findings to advance knowledge in cyberspace mental well-being and cyber threats.
- *Collaborate* with policymakers and industry leaders to develop strategies for a safer and more resilient digital environment.
- *Educate* the public on cyber threats, digital well-being, and the ethical use of science and technology for societal benefit.

Invited Talks, Keynotes, Research Presentations, and Field Activities

2026

- **Scientific Talk**, titled "Governance, Regulation, and Cyber Risk Management: Through the Business Lens," BEiNG-WISE COST Action Training School, University of Split, Croatia, September 7–11.
- **Panel Discussion** on "Shared Cybersecurity Paradigm: Aviation Flight vs. Health Care" conducted at Linköping Konsert & Kongress as part of the International Conference on Availability, Reliability and Security (ARES 2026), Linköping, Sweden, August 25.

2025

- **Organisation and Management** of "Capture the Flag (CTF) Competition" at Linköping University, Sweden, attracting over 120 participants from academia and industry, and supported through funding and sponsorships successfully secured from IEEE and several organisations, November 1.
- **Panel Discussion** on "Smart Space Aviation Technologies: Securing the Future Above Us" conducted at Vilnius Gediminas Technical University during the IEEE Nordic Young Professionals Congress, Vilnius, Lithuania, September 20.
- **Validation Exercise**, titled "SEC-AIRSPACE: Assessing and Enhancing Cybersecurity Awareness among Air Traffic Controllers," conducted at LFV Air Traffic Control Centre (ATCC), Malmö, Sweden, February 3–7.
- **Scientific Talk**, titled "Cybersecurity for Business," BEiNG-WISE COST Action Meeting, University of Calabria, Italy, June 23–24.
- **Conference Paper Presentation**, titled "On the Role of SecRAM Catalogues in ATM Cyber Risk Assessment and Improvement Opportunities," at the 25th Integrated Communications, Navigation, and Surveillance Conference (ICNS), Eurocontrol Headquarters, Brussels, Belgium, April 9.
- **Project Progress Presentation**, on the development of the module "Cybersecurity for Business" and the use of modern pedagogical tools, EU-funded Digital4Business (D4B) Project meeting, University of Bologna, Italy, February 26–27.
- **Scientific Talk**, titled "Bridging the Gaps in SecRAM 2.0: Framework Improvements and Development of a Dynamic Risk Assessment Tool," BEiNG-WISE COST Action Meeting, University College Dublin (UCD), Ireland, January 15–16.

2024

- **Technical Briefing**, to the Aviation Portfolio Council, Trafikverket, Visualization Center, Norrköping, Sweden, August 21.
- **Scientific Talk**, titled "Security and Privacy in the Future Communication Infrastructure of Aviation," BEiNG-WISE COST Action Meeting, University Ss. Cyril and Methodius in Skopje, North Macedonia, June 24–25.
- **Research Presentation**, titled "Progress on SecRAM Revisions and Development of a Cyber-Attack Integrated Simulator for Training Aviation Stakeholders," SESAR JU–funded SEC-AIRSPACE Project consortium meeting, Cefriel, Milan, Italy, March 14–15.

2023

- **Conference Paper Presentation**, titled "Enhancing Cybersecurity for LDACS: A Secure and Lightweight Mutual Authentication and Key Agreement Protocol," at the IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC), Barcelona, Spain, October 1–5.
- **Expert Talk**, titled "Vulnerabilities in Aviation Data Links: A Deep Dive into CPDLC and LDACS," Linköping University (LiU), Sweden, organized by Luftfartsverket, April 26.
- **Keynote Speaker**, titled "Securing the Industrial Landscape: Navigating Challenges and Opportunities in Cybersecurity," at the 1st International Conference on Artificial Intelligence and Green Computing (ICAIGC 2023), organized by Data4Earth Laboratory, University Sultan Moulay Slimane, Morocco, March 16.
- **Invited Speaker**, titled "Cybersecurity from an Industrial Context: Challenges and Opportunities," Uka Tarsadia University, India, organized by BlockSecAI Club (Webinar), February 4.

Invited Talks, Keynotes, Research Presentations, and Field Activities (continued)

2022 **Guest Talk**, titled "Cyber Security in Aviation: Threats, Challenges, and Research Directions," Indian Institute of Technology (IIT), Tirupati, India, November 17.

Mentoring and Supervision

Ph.D. Co-Supervision (1)

2022–2026 **Suleman Khan**, "Secure Mobility and Authentication Protocols in Heterogeneous Aviation Data Networks", Linköping University, Sweden (funded by *WASP*, *Luftfartsverket*, and *Trafikverket*); recipient of the prestigious Lawson Award.

Master's Thesis Supervision (25)

2025 **Adrian Rosén**, "Achieving Zero Trust – A Strategic Roadmap for Phased Implementation of Zero Trust Architecture", Linköping University, Sweden (Funded by *Kriminalvården*, Sweden).

Patric Rönn, "Assessing and Improving the Security Posture of a Cloud-Based SaaS Application for SMEs", Linköping University, Sweden (Funded by *Sys Partner*, Sweden).

Martin Hägglund, "Investigating Starlink's Resilience to GPS Spoofing and Space Weather Threats: Assessing the Impact of Man-Made and Natural Disruptions on Satellite Communication Performance", Linköping University, Sweden.

2024 **Abdullah Bin Zubair and Nuha Batool Taqi**, "Design, Implement, and Evaluate the Performance of an IPsec-Inspired Security Framework for HIP-VPLS Environments", Linköping University, Sweden.

Jonathan Eshak, "How Does the Use of Autonomous Penetration Testing Strengthen The Continuous Integration Flow?", Linköping University, Sweden (Funded by *SysPartner*, Sweden).

Gabriel Hult, "Evaluation of Unsupervised Anomaly Detection in Structured API Logs – A Comparative Evaluation with a Focus on API Endpoints", Linköping University, Sweden (Funded by *Upsales*, Sweden).

2023 **Casper Jensen**, "Asserting Password Crackers' Ability to Target Swedish Passwords – An Analysis", Linköping University, Sweden (Funded by *Nationellt Forensiskt Centrum (NFC)*, Sweden).

Adil Imran and Mirna Ghazzawi, "Design and Implementation of an Efficient Intrusion Response System for 5G RAN Baseband Units", Linköping University, Sweden (Funded by *Ericsson*, Sweden).

Lukas Nee, "Peer-to-Peer Misbehaviour Reporting Using Non-Interactive Zero Knowledge Proofs in ETSI ITS-G5", Linköping University, Sweden (Funded by *KnowIT*, Sweden).

2022 **Hanna Gustafsson and Hanna Kvist**, "Cyber Security Demonstration Using Penetration Testing on Wi-Fi Cameras", Linköping University, Sweden (Funded by *Total Defense Research Institute (FOI)*, Sweden).

Adrian Byström and Mattias Salo, "An Evaluation of Express Data Path from a 5G Perspective", Linköping University, Sweden (Funded by *Ericsson*, Sweden).

Yousef Hashem and Elmedin Zildzic, "Endpoint Intrusion Detection and Response Agents in Embedded RAN Products: A suitability and performance evaluation", Linköping University, Sweden (Funded by *Ericsson*, Sweden).

2021 **Awantika Tyagi**, "Implementation of Security Protocol for Protecting Personal Voice Assistants", Lovely Professional University, India.

Mentoring and Supervision (continued)

- 2020 **Karanjeet Choudhary**, "Implementation of Authentication Protocol for Industry 4.0 Networks", Lovely Professional University, India.
- 2019 **Preksha Jain**, "Performance Characterization of OMA and NOMA Systems", Lovely Professional University, India.
- 2018 **Charanpreet Kaur**, "Design and Development of Authentication Protocol for D2D Communications", Lovely Professional University, India.
- **Valmiki Siddhartha**, "Design and Development of Authentication Protocol for Internet of Things", Lovely Professional University, India.
- 2017 **Thokchom Saroj**, "A Lightweight Authentication Protocol Based on ECC for Satellite Communications", Lovely Professional University, India.
- **Gurpreet Kaur**, "Implementation of Message Authentication Code Using DNA-Bernoulli Model and Novel Hash Algorithm", Lovely Professional University, India.
- 2016 **Vikas Dhawan**, "Enhancing Data Integrity in Wireless Sensor Networks: A Novel Cryptographic Approach", Lovely Professional University, India.
- **Priya Verma**, "Building Secure Wireless Sensor Networks for Data Sensitive Applications", Lovely Professional University, India.
- 2015 **Jatinder Kaur**, "Optimizing Energy Conservation in Sensor Nodes with Adaptive Node Clustering Mechanisms", Lovely Professional University, India.
- **Arun Gautam**, "Application of Chaotic Functions in Constructing Substitution Boxes for Cryptographic Algorithms", Lovely Professional University, India.
- 2014 **Gurkiran Kaur**, "Energy Conservation and Collision Avoidance in Wireless Sensor Networks Using Adaptive MAC Protocols", Lovely Professional University, India.
- 2013 **Amoldeep Singh**, "Cognitive Wireless Sensor Networks: Implementing Self-Learning Capabilities for Improved Performance", Lovely Professional University, India.

Travel Grants

- 2026 Participated in the **ENERGYCOM Network Meeting**, Kaunas University of Technology (KTU), Kaunas, Lithuania. *Funded by Nordplus Higher Education (NPHE-2026/10449)* | *SEK 8.8K*
- Participated in the **COST Action Management Committee Meeting** at the University of Split, Croatia. *Funded by COST Action BEiNG-WISE (CA22104)* | *SEK 12.5K*
- **Research Visit**, UiT The Arctic University of Norway, Narvik, Norway. *Funded by Nordplus Higher Education (NPHE-2025/10355)* | *SEK 8.8K*
- Participated in the **ENERGYCOM Network Meeting**, Tallinn University of Technology (TalTech), Tallinn, Estonia. *Funded by Nordplus Higher Education (NPHE-2025/10355)* | *SEK 10.7K*
- Participated in the **COST Action Management Committee Meeting** at the Sapienza University of Rome, Italy. *Funded by COST Action BEiNG WISE (CA22104)* | *SEK 10.5K*
- Attended the **Digital4Business Interim Project Review Meeting** at the European Commission headquarters in Belgium. *Funded by EU Digital4Business* | *SEK 11K*
- 2025 Attended the **Young Professionals Congress** in Vilnius, Lithuania. *Funded by IEEE Sweden Section* | *SEK 5K*
- Participated in the **Young Researcher Training** at the EU COST Headquarters, Brussels. *Funded by EU COST Academy* | *SEK 10K*
- Participated in the **COST Action Management Committee Meeting** at the University of Calabria, Italy. *Funded by COST Action BEiNG WISE (CA22104)* | *SEK 12K*

Travel Grants (continued)

- Attended the **Digital4Business Consortium Meeting** at the University of Bologna, Italy. *Funded by EU Digital4Business* | *SEK 11.5K*
- Participated in the **COST Action Management Committee Meeting** at the University College Dublin, Ireland. *Funded by COST Action BEiNG WISE (CA22104)* | *SEK 12.5K*

2024
- **Research visit (STSM)** at the University of Warwick, UK. *Funded by COST Action BEiNG WISE (CA22104)* | *EUR 2.23K*
- Attended the **Young Professionals Congress** in Grenoble, France. *Funded by IEEE Region 8 (EUR 350) & IEEE Sweden Section (EUR 800)*
- Participated in the **COST Action Management Committee Meeting** at the University Ss Cyril and Methodius, Skopje, N. Macedonia. *Funded by COST Act. BEiNG WISE (CA22104)* | *SEK 12K*
- Participated in the **COST Action Management Committee Meeting** at INRIA Lille, France. *Funded by COST Action BEiNG WISE (CA22104)* | *SEK 12K*
- Attended the **SEC-AIRSPACE Consortium and Research Progress Meeting** at Cefriel, Milan, Italy. *Funded by EU Horizon - SESAR JU* | *SEK 9K*

2023
- Attended the **Young Professionals Congress** in Krakow, Poland. *Funded by IEEE Sweden Section* | *SEK 6K*
- Participated in the **Kick-off Meeting** of COST Action BEiNG WISE (CA22104) at the COST Association Headquarters, Brussels, Belgium. *Funded by EU COST* | *SEK 13.5K*
- Attended the **42nd DASC Conference** at Universitat Politècnica de Catalunya, Barcelona, Spain. *Funded by SEC AIRSPACE (EU HORIZON SESAR JU)* | *SEK 20K*
- Attended **18th SNCNW Conference** at Kristianstad University, Kristianstad, Sweden. *Funded by ELLIIT, Sweden* | *SEK 5K*

2022
- Delivered a **guest talk** at IIT Tirupati, India. *Funded by Linköping University* | *INR 100K*

Editorial and Peer Review Activities

Conference and Workshop Roles

2026
- **Publicity Chair**, The International Conference on Availability, Reliability and Security (ARES), Linköping University, Linköping, Sweden, August 24–27.

2025
- **Session Chair & Reviewer**, The 25th Integrated Communications, Navigation, and Surveillance Conference (ICNS), EuroControl Headquarters, Brussels, Belgium, April 8–10.
- **Reviewer**, IEEE International Conference on Communications (ICC), Montreal, Canada, June 8–14.
- **Reviewer**, 30th European Symposium on Research in Computer Security (ESORICS), Université de Toulouse, France, September 22–26.

2024
- **Local Organizer**, 19th Swedish National Computer Networking and Cloud Computing Workshop (SNCNW), Sweden, June 11–12.

2022–2024
- **Technical Program Committee Member & Reviewer**, IEEE Global Communications Conference (GLOBECOM).

Editorial Roles

2024–2026
- **Guest Editor**, Special Issue on "Security and Privacy Challenges in IoT-Driven Smart Environments," MDPI Sensors.

Journal Reviewing

IEEE
- *Internet of Things*

Editorial and Peer Review Activities (continued)

- *Transactions on Dependable and Secure Computing (TDSC)*
- *Transactions on Information Forensics and Security (TIFS)*
- *Transactions on Network and Service Management (TNSM)*
- *Transactions on Intelligent Transportation Systems (T-ITS)*
- *Transactions on Industrial Informatics (TII)*
- *Transactions on Green Communications and Networking*
- *Network*

Elsevier
- *Computer Communications*
- *Vehicular Communications*

Springer
- *International Journal of Information Security (IJIS)*

Wiley
- *International Journal of Network Management (IJNM)*

References

Available on Request